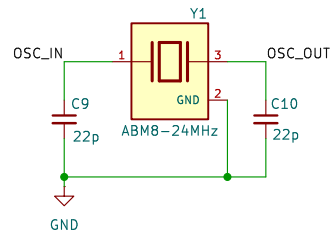


HSE CRYSTAL



Abrakon ABM8-24.000MHZ-B2T
24Mhz divides cleanly to 48 MHz USB clock and 170 MHz system clock.

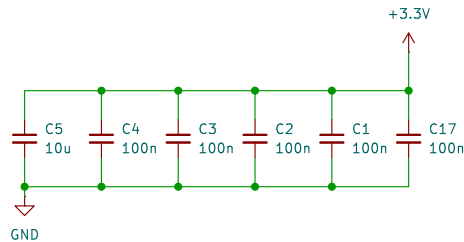
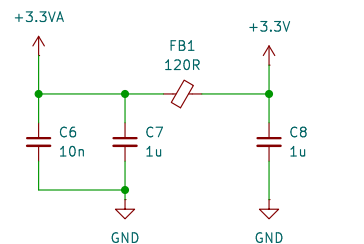
Capacitance Calculation:

$C_{load_target} = 18 \text{ pF}$ and $C_{stray} = 4 \text{ pF}$
 $C_{load} = (CL1 \times CL2) / (CL1 + CL2) + C_{stray}$
 $(CL1 \times CL2) / (CL1 + CL2) = 14 \text{ pF}$,
 if $CL1 = CL2$, each is 28 pF .

So 27 pF or 33 pF Capacitors
Most designs end up at 18–22 pF caps for an 18 pF crystal.

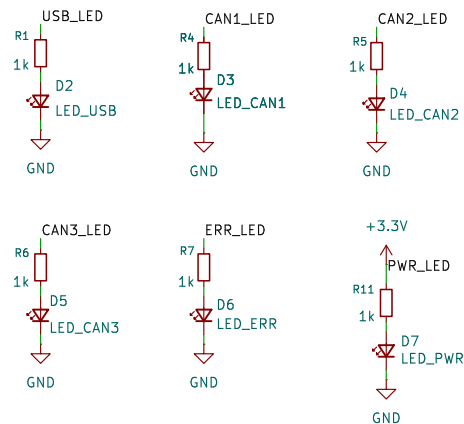
Tune this on the bench by trying 18 pF, 22 pF, 27 pF caps.

MCU DECOUPLING

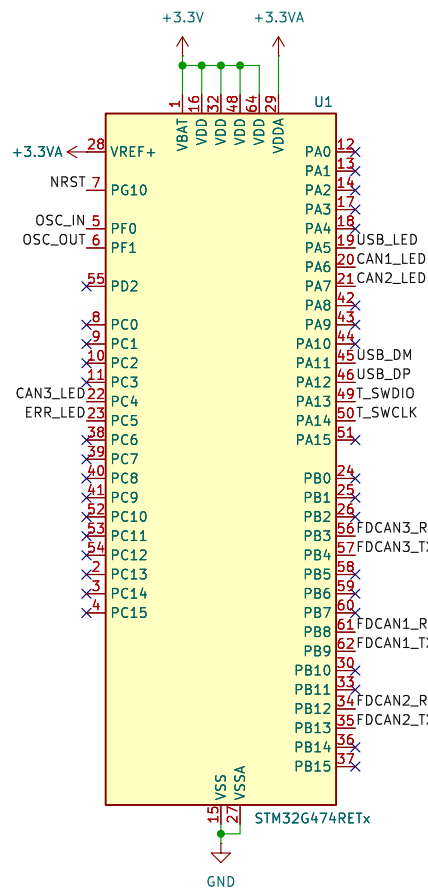


4x 100n for each VDD + 1x 100n for VBAT + 1x 10u Bulk

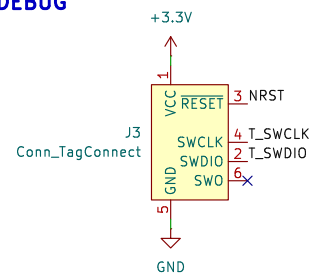
LED INDICATORS



MCU

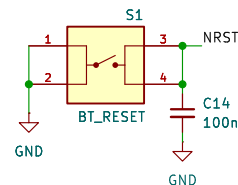


SWD DEBUG

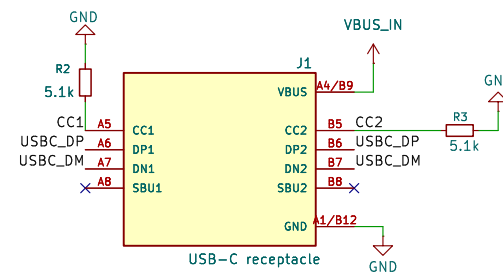


| Tag-Connect TC2030-IDC – No Legs

RESET SWITCH

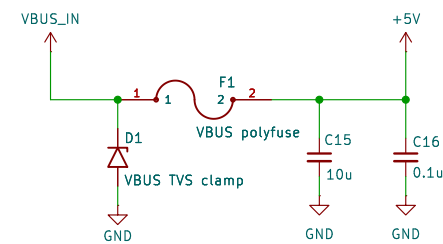


USB-C RECEPTACLE



USB-C Receptacle (GCT USB4110-GF-A)

VBUS PROTECTION

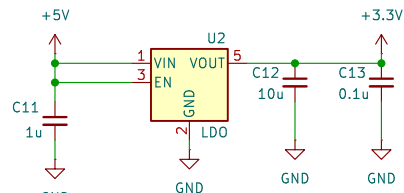


VBUS TVS Clamp (Littelfuse SMAJ5.0A)
Overvoltage protection

VBUS Polyfuse (Bourns MF-MSMF050-2)
500 mA hold, 1 A trip – Resettable overcurrent protection

1x 10u for Bulk + 1x 0.1u for Hi-Freq

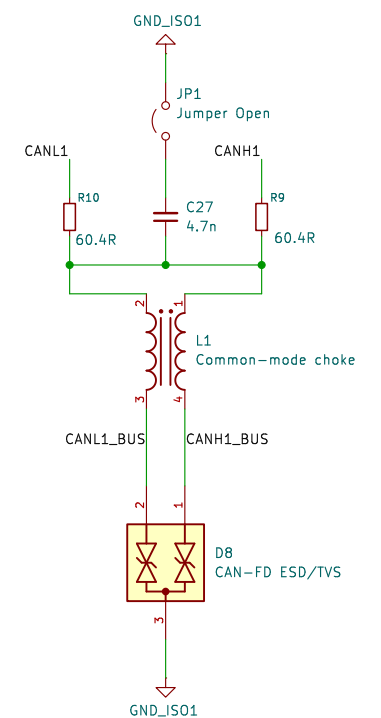
LDO POWER SUPPLY



LDO:
AP2112K-3.3 (SOT-25, 600 mA, 250 mV dropout)

- Total 3.3V load is ~100 mA
- LDO dissipation $(5.0 - 3.3) \times 0.1 = 170 \text{ mW}$

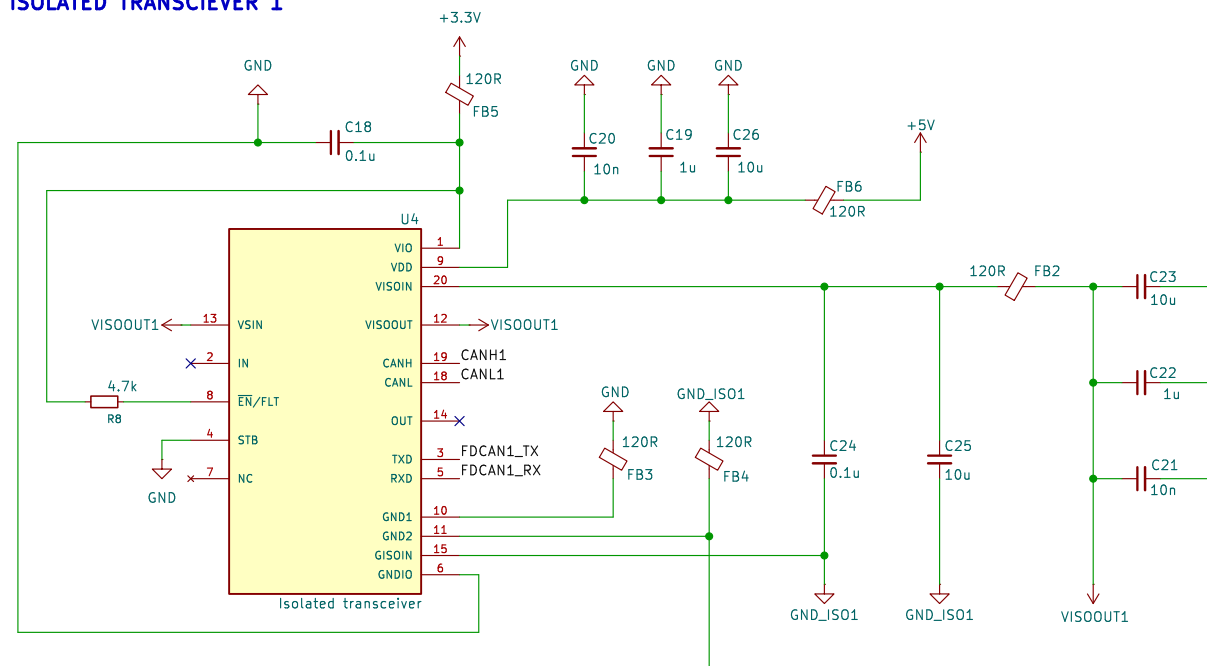
CAN TERMINATION & ESD



Common-mode choke (Würth 744232511)
Filters EMI and High-Freq Noise right after Termination

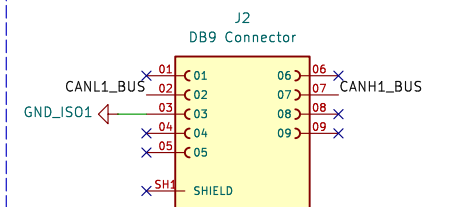
CAN-FD ESD/TVS (ESD2CAN24DBZRQ1)
Protects against ESD and Transients
Extra-low capacitance tuned for FDCAN speeds up to 5Mbps

ISOLATED TRANSCIEVER 1



ISOW1044 Isolated CAN FD Transceiver with Integrated Low-Emissions, Low-Noise, High-Efficiency DC-DC Converter

DB9 CONNECTOR 1



Charlie Shields

Sheet: /
File: CAN_KiCad.kicad_sch

Title: CAN Adapter V1

Size: A3	Date: 2026-04-26
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Size: A3	
KiCad E.D.A. 9.0.8	

Rev: 1/1